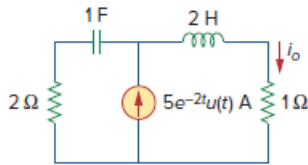


Problem

Determine $i_o(t)$ in the circuit in Fig. 16.69.

Figure 16.69:



Step-by-step solution

Step 1 of 4

Refer to the circuit in Figure 16.69 in the textbook.

Transform the circuit into s -domain.

$$5e^{-2t}u(t) \Rightarrow \frac{5}{s+2}$$

$$1 \text{ F} \Rightarrow \frac{1}{sC} = \frac{1}{s}$$

$$2 \text{ H} \Rightarrow sL = 2s$$

Draw the modified circuit as shown in Figure 1.

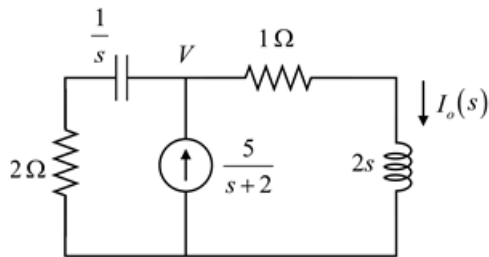


Figure 1

Step 2 of 4

The current $I_o(s)$ in the circuit is,

$$\begin{aligned} I_o(s) &= \left(\frac{5}{s+2} \right) \frac{2 + \frac{1}{s}}{\left(2 + \frac{1}{s} + 1 + 2s \right)} \\ &= \left(\frac{5}{s+2} \right) \frac{2s+1}{(2s^2+3s+1)} \\ &= \left(\frac{5}{s+2} \right) \frac{2s+1}{(2s+1)(s+1)} \\ &= \frac{5}{(s+2)(s+1)} \end{aligned}$$

Step 3 of 4

Apply partial fractions.

$$I_o(s) = \frac{A}{s+1} + \frac{B}{s+2} \dots\dots (1)$$

$$\frac{5}{(s+1)(s+2)} = \frac{A(s+2) + B(s+1)}{(s+1)(s+2)}$$

$$5 = A(s+2) + B(s+1)$$

$$5 = s(A+B) + (2A+B) \dots\dots (2)$$

Comparing s-coefficients in equation (2),

$$A + B = 0$$

$$A = -B \dots\dots (3)$$

Comparing constant coefficients in equation (2),

$$2A + B = 5$$

Substitute $-B$ for A.

$$2(-B) + B = 5$$

$$B = -5$$

Substitute -5 for B.

$$\begin{aligned} A &= -(-5) \\ &= 5 \end{aligned}$$

Step 4 of 4

Substitute 5 for A and -5 for B in equation (1).

$$I_o(s) = \frac{5}{s+1} - \frac{5}{s+2}$$

Apply inverse Laplace transform.

$$\begin{aligned} i_o(t) &= 5e^{-t}u(t) - 5e^{-2t}u(t) \\ &= 5(e^{-t} - e^{-2t})u(t) \text{ A} \end{aligned}$$

Thus, the current $i_o(t)$ is $\boxed{5(e^{-t} - e^{-2t})u(t) \text{ A}}$.